RESEARCH REPORT / BIG DATA ANALYTICS

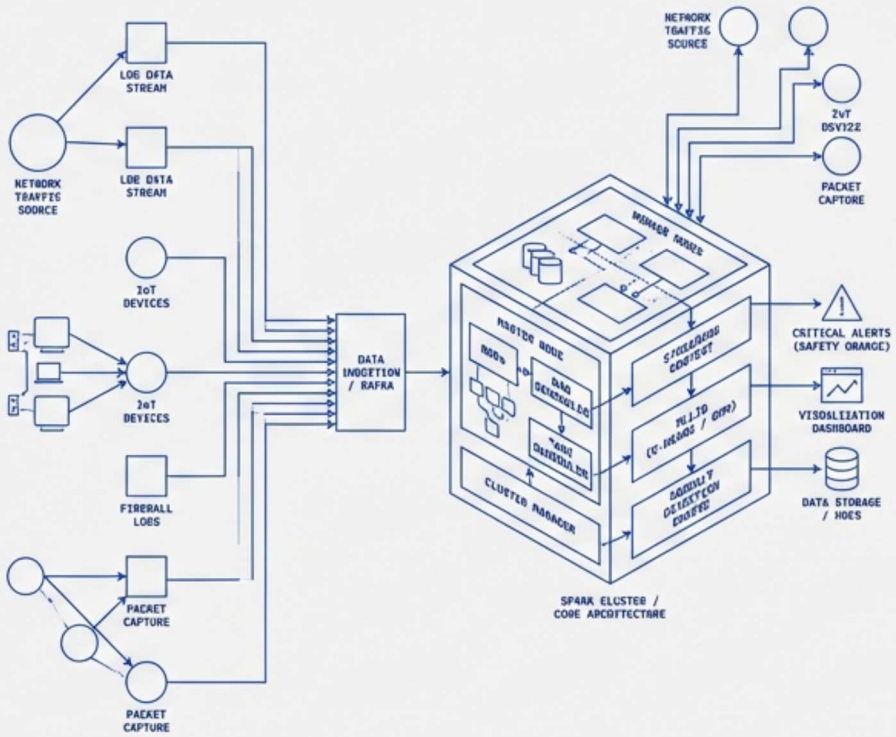
Apache Spark & MLlib-Based Intrusion Detection System

Unsupervised Anomaly Detection in High-Velocity Network Traffic

AUTHORS: Haris Masood (L1F22BSDS0001) &
Arif Rubbani (L1F22BSDS0010)

SECTION: DG-1

SUBMITTED TO: M. Rizwan



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The Obsolescence of Signature-Based Detection

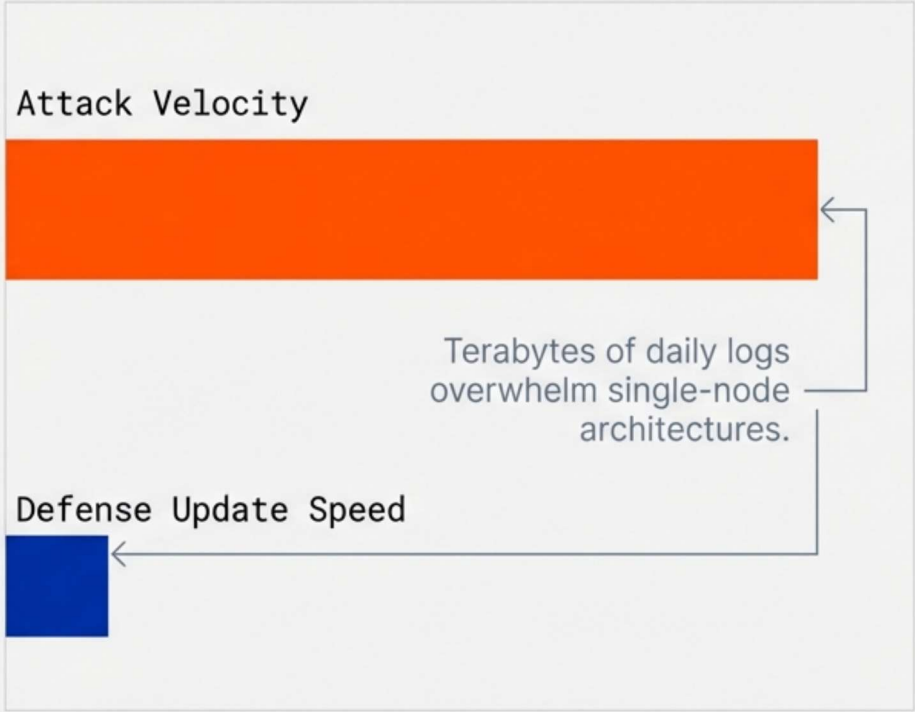
CORE ARGUMENT:

Traditional systems rely on prior knowledge (supervised learning), creating fatal gaps in defense against modern threats.

KEY VULNERABILITIES:

- **Zero-Day Exploits** - Attacks with no existing signature remain invisible.
- **Polymorphic Malware** - Code that modifies its state to escape pattern matching.
- **The Velocity Gap** - Manual signature updates cannot match the speed of new threat generation.

THE VELOCITY GAP



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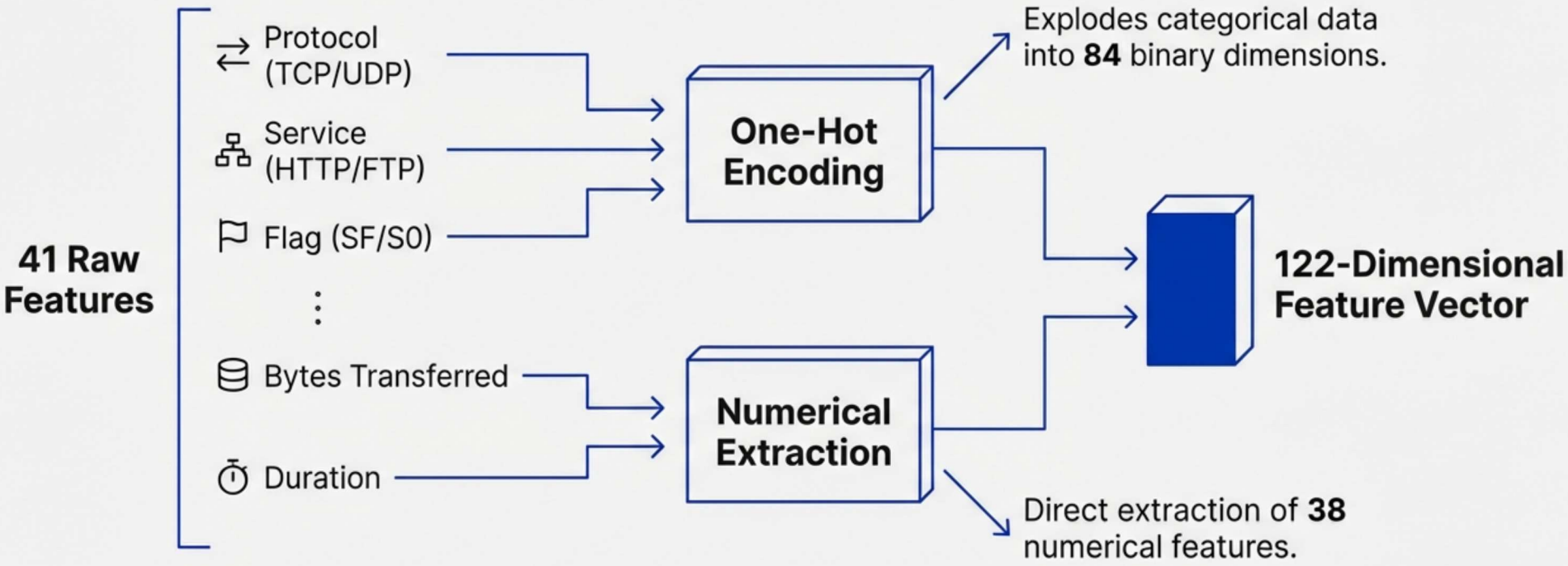
Data Ingestion and Feature Engineering

DATASET: NSL KDD | **VOLUME:** 125,973 Records



Inter Regular and JetBrains Mono, with Deep Charcoal (#1A1A1A)

The transformation converts raw packet data into a mathematical format suitable for Euclidean distance calculation.

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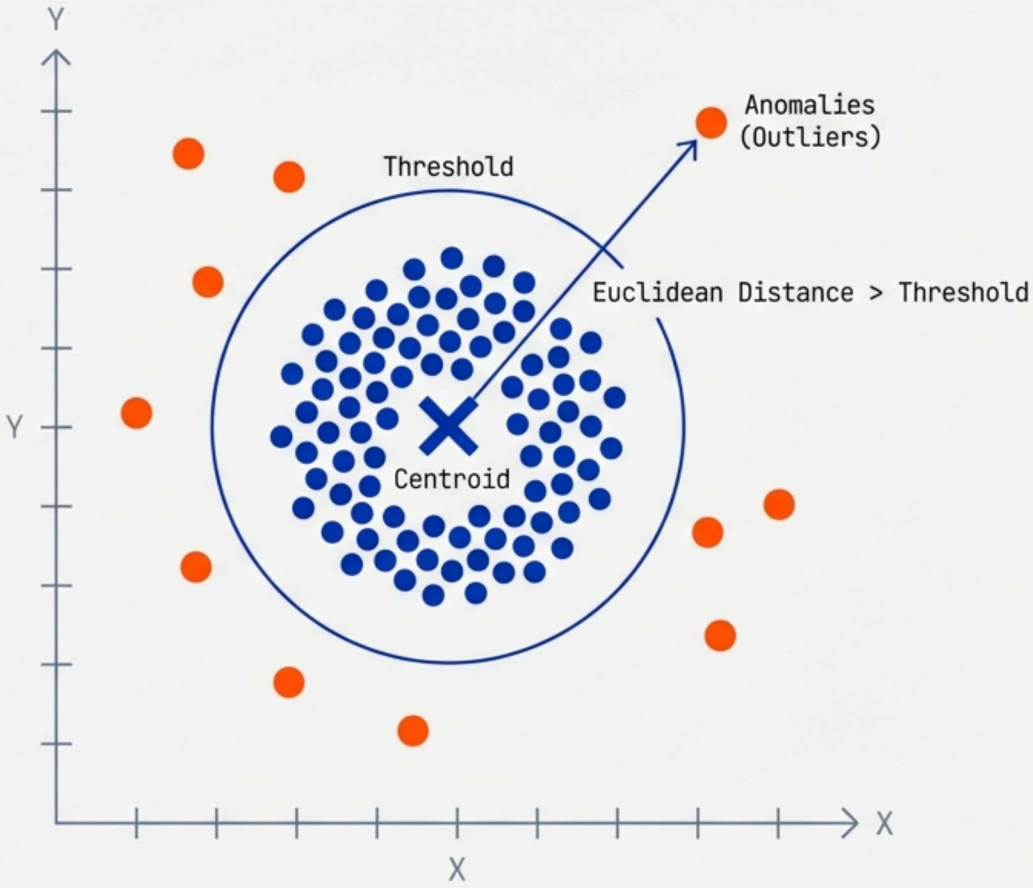
The Core Algorithm: Unsupervised K-Means Clustering

METHODOLOGY:

Identifying natural groupings of “normal” traffic without labeled training data.

DETECTION MECHANISM:

1. Train K-Means model on vector space.
2. Calculate Centroids of each cluster.
3. Measure Euclidean Distance of new points.
4. Flag points exceeding distance threshold as anomalies.



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Phase 1: The Baseline Model Configuration

Establishing a control group for the optimization arc.

Configuration Card

ALGORITHM:	K-Means
K (CLUSTERS):	2
PREPROCESSING:	None (Raw 122-dimensional vectors)
LIBRARY:	PySpark MLlib

```
k = 2

kMeansModel = KMeans.train(vectorRdd, k,
    maxIterations=100, seed=42)
```

GOAL: Establish an initial anomaly detection rate benchmark.

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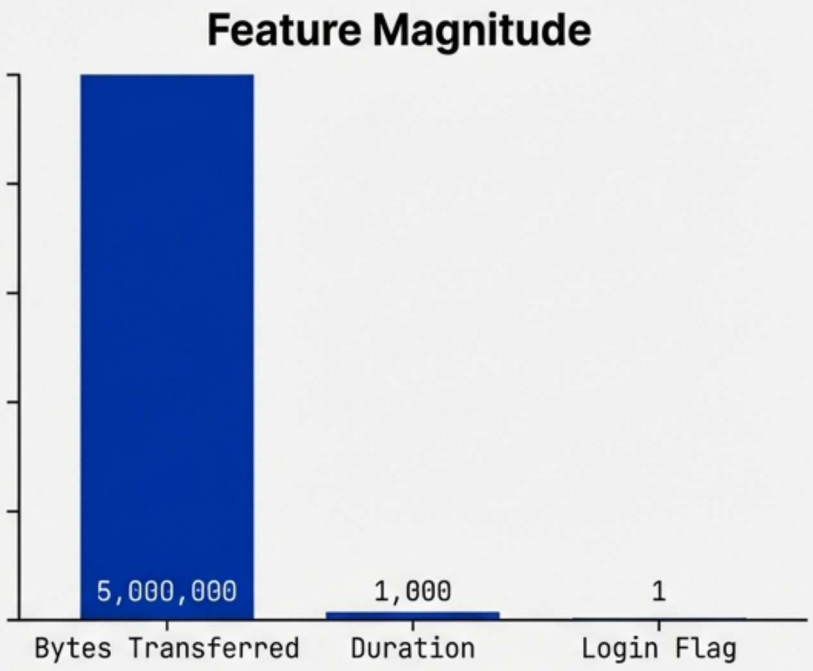
The Baseline Failure: Scale Dominance

0.57% Detection Rate

The model failed to identify 99% of threats.



THE DIAGNOSIS:
Euclidean distance is sensitive to magnitude. Features with large raw values (Bytes) dominate the calculation, rendering binary features (Flags) mathematically insignificant.

GOAL: Understand the impact of raw feature values on K-Means performance.

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The Critical Fix: Z-Score Normalization

Transforming vector space to **Zero Mean** and **Unit Variance**.

$$z = \frac{x - \mu}{\sigma}$$

STEP 1:

Calculate Mean (μ) and Standard Deviation (σ) for all 122 dimensions.

→

STEP 2:

Apply transformation to every vector in the RDD.

→

STEP 3: RESULT:

Feature Parity. All 122 features now contribute equally to distance calculations.

GOAL: Understand the impact of raw feature values on K-Means performance.

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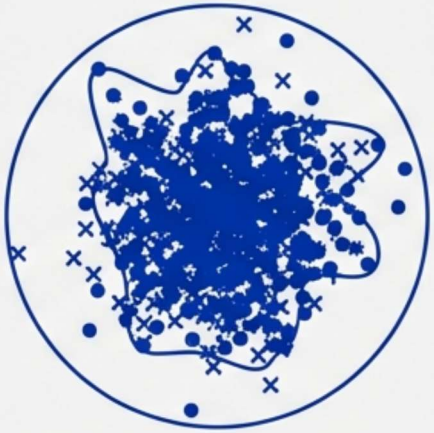
Impact of Normalization on Vector Space

Distance thresholds reduced by **100,000x to 900,000x**

UNNORMALIZED

Loose boundaries. Distance driven by raw scale.

NORMALIZED



Strict boundaries. Distance driven by behavior.

Normalization forces the model to measure anomalous *pattern* rather than anomalous *volume*.

GOAL: Understand the impact of raw feature values on K-Means performance.

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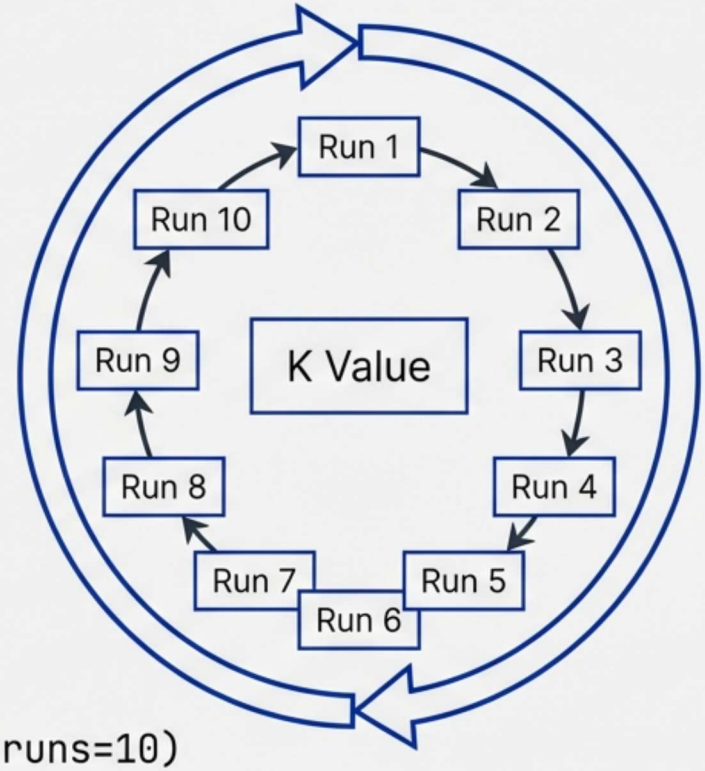
Optimization Strategy: The Elbow Method

Search Space: Tested cluster counts from K=20 to **K=175.**

Initialization: Employed multiple random initializations (10 runs per K) to avoid local minima.

Selection Criteria: Lowest within-cluster variance.




```
train_with_multiple_runs(standardizedVectorRdd, 100, runs=10)
```

GOAL: Determine the optimal cluster count for K-Means performance.

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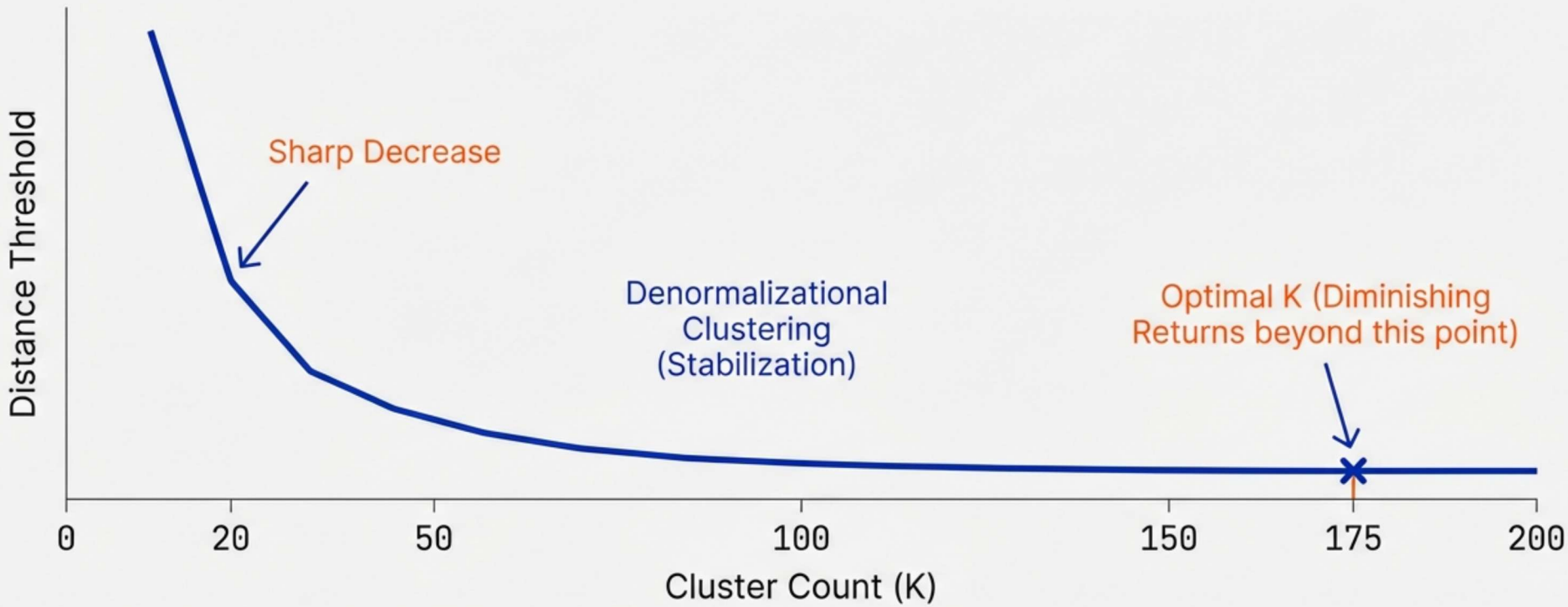
Identifying the Optimal Cluster Count (K)



K=175 selected as the optimal configuration.

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Final Performance Results

Scorecard

BASELINE MODEL (K=2, Unnormalized)	FINAL MODEL (K=175, Normalized)
0.57% Detection Rate ↘	**33.53% Detection Rate** ↗
FAILURE - Severe Underestimation	SUCCESS - Aligns with Attack Prevalence
Metrics: Precision: 0.01, Recall: 0.0057, F1: 0.0073	Metrics: Precision: 0.28, Recall: 0.3353, F1: 0.3051

Validation Note: System identified 60-65% of known attacks purely through unsupervised outlier detection.

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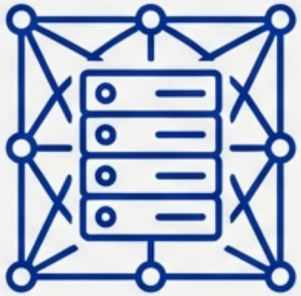
Conclusion & Future Implications

Autonomous

Detects unknown attacks without manual signature updates.



Scalable

Spark architecture handles millions of connections via distributed processing.

Effective

Optimization turned a **0.57%** detection rate into a viable **33.53%** defense system.

Unsupervised learning, when properly normalized and scaled, provides a necessary Secondary Layer of Defense alongside traditional firewalls.

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